Question 3 a

```
install.packages("GGally")
```

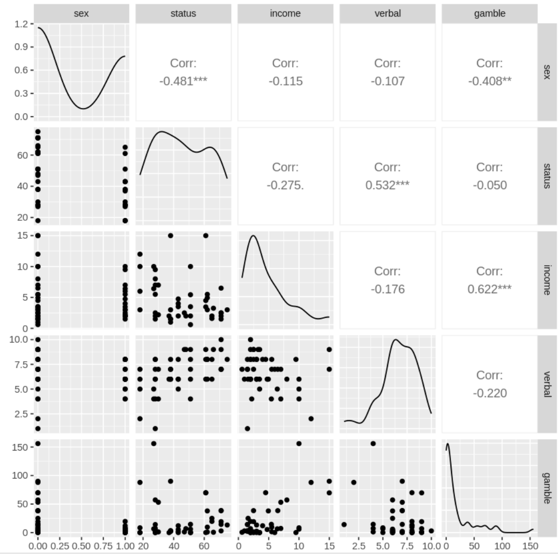
```
Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

also installing the dependencies ‘patchwork’, ‘ggstats’, ‘plyr’
```

```
install.packages("faraway")
```

```
Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

also installing the dependencies ‘minqa’, ‘nloptr’, ‘RcppEigen’, ‘lme4’
```

```
# Load necessary libraries
library(pander)
library(ggplot2)
library(GGally)
library(faraway)

# Load the dataset
data(teengamb, package="faraway")

# Plot pairwise relationships
ggpairs(teengamb)

# Fit the linear model
lm.fit <- lm(gamble ~ sex + status + income + verbal, data=teengamb)

# Display model summary
summary(lm.fit)

# Print regression coefficients
coefficients <- coef(lm.fit)
print(coefficients)
```

```
Attaching package: ‘faraway’


The following object is masked from ‘package:GGally’:

    happy



Call:
lm(formula = gamble ~ sex + status + income + verbal, data = teengamb)

Residuals:
    Min      1Q  Median      3Q     Max 
-51.082 -11.320  -1.451   9.452  94.252 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  22.55565   17.19680   1.312   0.1968    
sex         -22.11833    8.21111  -2.694   0.0101 *  
status        0.05223    0.28111   0.186   0.8535    
income        4.96198    1.02539   4.839 1.79e-05 ***
verbal       -2.95949    2.17215  -1.362   0.1803    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 22.69 on 42 degrees of freedom
Multiple R-squared:  0.5267,	Adjusted R-squared:  0.4816 
F-statistic: 11.69 on 4 and 42 DF,  p-value: 1.815e-06
 (Intercept)          sex       status       income       verbal 
 22.55565063 -22.11833009   0.05223384   4.96197922  -2.95949350 
```



Question 3 b

```
# Fitted values and residuals from the model
fitted_values <- fitted(lm.fit)
residuals <- residuals(lm.fit)

# Sum of squares of residuals (SSR)
ssr <- sum(residuals^2)

# Sample size (n) and number of predictors (p)
n <- length(residuals)
p <- length(coef(lm.fit))

# Unbiased estimate of variance
unbiased_est <- ssr / (n - p)
print(unbiased_est)
```

```
[1] 514.8516
```

Question 3 c

```
# Proportion of variance explained by the predictors
var.explained.proportion <- summary(lm.fit)$r.squared
var.unexplained.proportion <- 1 - summary(lm.fit)$r.squared
```

```
# Display proportion of variance explained
pander(data.frame(var.explained.proportion= var.explained.proportion), caption="Proportion of Variance Explained")
```

```
| var.explained.proportion |
|:------------------------:|
|          0.5267          |

Table: Proportion of Variance Explained
```

Question 3 d // the sample correlation between the residuals and the fitted values in regression models is always zero, this is due to minimizing sume of squared residuals.

Question 3 e Zero correlation, if Income is a predictor it is already accoutned by model.

Question 3f

Simple Explanation:

beta 1 (coefficient for the variable "sex") will tell us how much more or less males spend on gambling compared to females. if beta 1 is greater than zero, males spend more than female and vice versa

Question 4

```
# Load necessary data and libraries
data(prostate, package="faraway")
library(broom)

# Create a dataframe to hold the results of the experiment
model.stats <- data.frame(num.predictors=integer(), r.squared=numeric(), residual.se=numeric(), model.string=character())

# Fit the first model with 1 predictor
lm.fit <- lm(lpsa ~ lcavol, data=prostate)
summary(lm.fit)
model.summary <- summary(lm.fit)

# Extract statistics
r.squared <-  model.summary$r.squared
residual.se <- model.summary$sigma
model.string <- "lpsa ~ lcavol"

# Append results to the dataframe
model.stats <- rbind(model.stats, list(num.predictors = 1, r.squared=r.squared, residual.se=residual.se, model.string=model.s

# Fit models with increasing number of predictors
predictor.combinations <- list(
  "lpsa ~ lcavol + lweight",
  "lpsa ~ lcavol + lweight + svi",
  "lpsa ~ lcavol + lweight + svi + lbph",
  "lpsa ~ lcavol + lweight + svi + lbph + age",
  "lpsa ~ lcavol + lweight + svi + lbph + age + lcp",
  "lpsa ~ lcavol + lweight + svi + lbph + age + lcp + pgg45",
  "lpsa ~ lcavol + lweight + svi + lbph + age + lcp + pgg45 + gleason"
)

# Loop through each model combination
for (i in seq_along(predictor.combinations)) {
  lm.fit <- lm(as.formula(predictor.combinations[[i]]), data=prostate)
  model.summary <- summary(lm.fit)

  # Extract statistics
  r.squared <-  model.summary$r.squared
  residual.se <- model.summary$sigma
  model.string <- predictor.combinations[[i]]

  # Append results to the dataframe
  model.stats <- rbind(model.stats, list(num.predictors = i + 1, r.squared = r.squared, residual.se = residual.se, model.stri
}

# Convert the model strings to character
model.stats$model.string <- as.character(model.stats$model.string)

# Load pander for table display
install.packages("pander")
library(pander)
```

```
library(pander)

# Display the model stats
rownames(model.stats) <- NULL
pander(model.stats, caption = "Model Stats")

# Plot the relationship between Residual SE and R-squared
install.packages("ggplot2")
install.packages("GGally")
library(ggplot2)
library(GGally)

p <- ggplot(model.stats, aes(x = residual.se, y = r.squared)) + geom_point()
p <- p + geom_text(aes(label = num.predictors, hjust = 1.5, vjust = 1.5))
p + ggtitle("Residual SE vs R-squared with Number of Predictors") + theme(plot.title = element_text(hjust = 0.5))
```

```
Call:
```